NEWSLETTER

DCSA CONNECT

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DEPARTMENT OF COMPUTER SCIENCE & APPLICATIONS

PANJAB UNIVERSITY, CHANDIGARH

IN THIS EDITION

YOU'LL KNOW ABOUT

03 • **DEPARTMENT EVENTS**
Event highlights of the month

05 • **PLACEMENT UPDATES**
Recent updates related to placements

06 • **FACULTY OUTREACH**
Faculty Collaboration and Engagement

07 • **ALUMNI CORNER**
A word from our alumni & their achievements

10 • **RESEARCH CORNER**
Highlights of research work

18 • **ABOUT FACULTY**
Get to know about our faculty

19 • **THE EDITORIAL TEAM**
Meet our editorial team

TPC Team and Editorial team Felicitation



The Felicitation Ceremony to bid farewell to the Training and Placement Cell (TPC) and the Editorial Team took place on December 23, 2024, graced by the presence of the Chairperson, Dr. Anuj Sharma, along with Dr. Jasleen Kaur Bains, Mrs. Supreet Thapar, Dr. Kavita Taneja, and Dr. Rohini Sharma. This event celebrated the remarkable dedication and contributions of both teams toward the growth and success of the department.

The ceremony commenced with an inspiring address by Dr. Anuj Sharma, who welcomed the attendees and highlighted the significant efforts and accomplishments of the team members over the year. Following this,the current TPO, Dr. Jasleen Kaur Bains, addressed the gathering by sharing her experiences and learnings from the placement process and expressed heartfelt appreciation for the outgoing members and offered encouragement to the incoming team set to take charge in January.

Dr. Kavita Taneja and Mrs. Supreet Thapar also shared their words of gratitude and wisdom, applauding the teams and imparting valuable insights to the students The event progressed with speeches from the heads of both teams, who reflected on their personal journeys, challenges faced, and key moments of the past year. These reflections served as a source of motivation and guidance for the incoming team members.

Vaibhav Sharma, Chief Coordinator of the Training and Placement Cell, elaborated on his interactions with HR professionals from various companies.

He shared strategies for effectively engaging them, attracting placements to the campus, and maximizing opportunities for students. Similarly, Gurbir Singh Dhaliwal, Editorial Head of the Newsletter Team, offered insights into his unique storytelling approach, emphasizing the importance of presenting departmental activities authentically to make every reader feel acknowledged and engaged.

The highlight of the ceremony was the distribution of certificates and mementos, recognizing the efforts and achievements of all team members. Additionally, special appreciation certificates were presented by the Editorial Team to Team Coordinators Dr. Jasleen Kaur Bains and Mrs. Supreet Thapar, along with Research Scholar S. Ratna, in acknowledgment of their exceptional support.

The event concluded with a joyous cake-cutting ceremony, symbolizing the unity and achievements of both the TPC and the Editorial Team. Dr. Anuj Kumar and Dr. Kavita Taneja will be taking charge of the TPC team as new coordinators in 2025. This memorable occasion, filled with gratitude, inspiration, and a sense of accomplishment, left everyone motivated and eager to achieve even greater milestones in the future.

- Sargun and Arsh Divya MCA-I(M)

December Exams: A Milestone For MCA & M.Sc. Students

The December odd semester examinations were conducted successfully for all academic programs, including MCA (I & II) and M.Sc.(I & II). These exams marked significant milestones for students, testing their understanding and preparation for their respective courses.
Exams were conducted under the supervision of Centre Superintendent Dr. Jasleen Kaur Bains and Deputy Superintendent Dr. Rohini Sharma from **December 3, 2024 to December 31, 2024.**

- Nancy MCA-I(E)

Placement Updates

Virtualise Technology Placement Drive

The Department conducted a successful placement drive in collaboration with Virtualise Technology, aiming to provide students with a platform to launch their professional careers. The placement process commenced on December 11, 2024, with an initial technical evaluation focused on testing the students' proficiency, critical thinking, and subject mastery.

A total of 18 students from various academic programs participated in the test, demonstrating their preparedness to tackle real-world challenges. Based on their performance, five students were shortlisted for the interview round held on December 13, 2024. After a comprehensive evaluation during the interviews, Aakash and Varinder from MCA-II (E) emerged as the final selections for the role of Software Engineer Interns. Their internships with Virtualise Technology are scheduled to commence in January 2025.

- Arsh Divya MCA-I(M)

Keymouse IT Placement Drive

The Department successfully organized a placement drive in collaboration with KeymouseIT, providing students with significant career opportunities. The placement drive commenced with a technical test and initial interviews conducted on December 12, 2024, designed to evaluate candidates' technical expertise and problem-solving abilities. After rigorous assessment, a total of 7 students were shortlisted for the next stage.

These shortlisted candidates were invited to the KeymouseIT office for a final interview round with the CEO. This crucial round focused on assessing their interpersonal skills, confidence, and overall suitability for the roles. Based on their stellar performance, four students—Ajay, Aakash, Varinder from MCA-II (E) and Sneha Goyal from MCA -Hoshiarpur campus were selected.

- Kashish Dhamija MCA-I(M)

Dr. Anuj Kumar Graces Himachal Mahasabha Cricket Tournament



Dr. Anuj Kumar was invited as the Chief Guest for a cricket tournament organized on December 9, 2024 by the Himachal Mahasabha in Chandigarh. Upon arrival, Dr. Anuj Kumar was warmly welcomed and had the honor of cutting the ribbon to officially inaugurate the event. In his address, he emphasized the importance of sports in fostering teamwork, discipline, and a healthy society.

He encouraged participants to balance academics with extracurricular activities like sports. The tournament featured exciting matches, and Dr. Anuj Kumar interacted with players, offering motivation and support. At the closing ceremony, he presented trophies to the winning team and congratulated all participants for their enthusiasm. His presence added immense value to the event, reinforcing the significance of community engagement and sports in personal and social development.

- Ishneet Kaur MCA-I(M)

Alumni Meet 2024 - DCSA, Panjab University



DCSA Alumni Meet, held on December 21, 2024, was a momentous occasion, reuniting esteemed alumni, faculty, staff, and students to commemorate DCSA's illustrious legacy. The event commenced with a traditional lamp lighting ceremony, accompanied by soulful instrumental music, setting the tone for a nostalgic evening. The ceremony was followed by Panjab University Anthem, bringing everyone together in unison. A two-minute silence was observed in remembrance of late Dr. Sudhir Rohilla, the first JRF of DCSA.

Floral welcomes were offered to the Silver Jubilee alumni, including Ms. Neetu Gupta, Mr. Narinder Kamra, Dr. Mandeep Walia, Ms. Archana, Ms. Monika Aggarwal, and Mr. Sanjay, by Chairperson and faculty. The welcome addresses by Dr. Anuj Sharma (Chairperson) and Prof. Anu Gupta (Alumni Coordinator) warmly acknowledged the gathering, highlighting the significance of alumni connections. The Alumni Keynote Address was delivered by Dr. Mandeep Walia, who shared valuable insights from his distinguished IT career and leadership journey.

The event then proceeded with the introduction of Faculty, Librarian, Technical Staff, Office Staff, and Helping Staff, acknowledging their continuous contributions to the department. The captivating DCSA AV film took the audience on a journey down memory lane, revisiting milestones and achievements. The nostalgia segment stole the show with alumni sharing heartfelt experiences, including doctoral alumni and silver jubilee

alumni, who fondly reminisced about their time at DCSA, reliving cherished memories and milestones.

The audience was mesmerized by performances of students, alumni, and faculty - featuring singing, beatboxing, creative poetry recital, and dance act. The event concluded with alumni felicitation and a heartfelt vote of thanks by Ms. Supreet (Alumni Co-Coordinator), followed by High Tea, providing a final opportunity for networking and reminiscing. This memorable gathering not only celebrated the achievements of the silver jubilee batch but also strengthened the bond between alumni, faculty, and staff, marking an important milestone in the history of DCSA, Panjab University.

-Jaskaran & Ishneet Kaur MCA-I(M)

A Tribute to the Life and Legacy of Dr. Sudhir Rohilla





Dr. Sudhir Rohilla, the department's first Junior Research Fellow (JRF) and an esteemed alumnus who pursued his PhD (2013–2018), was known for his brilliance, passion for education, politeness, and deep kindness. He was selected for the prestigious Biometric Society workshop in Italy, a fully funded honor that recognized his potential and contributions to the field. Additionally, he worked as a Project Scientist-1 at CSIR-IGIB (Institute of Genomics and Integrative Biology), New Delhi, where his research left a lasting impact.

While his academic achievements were remarkable, it was his respectful and gentle nature, combined with his sharp analytical mind, that made him truly beloved by his peers and students. As an Assistant Professor at Gopichand Arya Mahila College, Abohar, Dr. Rohilla inspired countless students with his dedication to their growth and his unwavering commitment to their success.

Tragically, Dr. Rohilla passed away on December 9th, leaving behind a legacy of kindness, insight, and a vision that continues to inspire. Dr. Rohilla's memory will remain etched in our hearts. Though he is no longer with us, the values he upheld and the vision he passionately shared will continue to guide and motivate us.

Dr.Sudhir Rohilla : Memories, Messages and Feedback





Favourite memories at DCSA:

"The evening tea times were truly special. We shared thoughts and ideas, met new people who became an integral part of our lives, and had meaningful conversations about each other's journeys. Those moments of connection and laughter made the experience unforgettable and enriched our journey together."

Feedback for the department:

"I suggest creating a university-wide computer society at Panjab University. This initiative would allow students from across the university to feel connected to DCSA and foster a sense of community. It could serve as a platform for collaboration, knowledge-sharing, and engagement with technology, benefiting both DCSA and the broader student body."

Message for Current Students:

"Congratulations on becoming part of our DCSA family! The road ahead is full of opportunities, so stay curious, keep experimenting, and don't be afraid to fail. Trust in your ability to learn and grow. Whether your journey leads to creating, discovering, or leading, remember to enjoy each step and celebrate your progress. The world needs your unique talents and ideas—go out there and make a difference!"

-S Ratna (Research scholar)

International Conference on "Computing Advances in Artificial Intelligence And Machine Learning" (ICCAAIML-2024)





The Department of Computer Science and Applications (DCSA) at Panjab University, Chandigarh, successfully hosted the International Conference on Computing Advances in Artificial Intelligence and Machine Learning (ICCAAIML) on December 16–17, 2024. This highly anticipated two-day event brought together a dynamic community of scholars, researchers, and industry professionals from around the world. ICCAAIML was a festival of research, offering a distinguished platform for sharing and exchanging research ideas,engaging in discussions on cutting-edge technology, and exploring the diverse fields of Artificial Intelligence(AI) and Machine Learning (ML).

The conference emphasized on advancing science and technology, and management through interdisciplinary and multidisciplinary research, fostering transformative solutions. The conference featured four thoughtfully curated tracks, each designed to delve into key areas of AI and ML research: AI and Deep Learning, Data Communication and Computer Networks, Deep Intelligence for Interdisciplinary Research and, Immersive Technologies. The conference was organized and led by the conference coordinators, Dr. Anuj Sharma, Chairperson of DCSA, and Prof. Sonal Chawla, with invaluable support from Prof. R.K. Singla, a senior and esteemed faculty member, along with other respected faculty members, staff, and enthusiastic participants, whose collective efforts made the event a great success.

Day 1 : Inauguration and Keynote Sessions, Book Launch and Research Presentations

The inaugural day began with the registration process, where participants warmly registered and received conference kits containing the schedule, brochures, and other essential materials. This was followed by the official opening ceremony, which featured the lighting of the lamp, a sacred symbol of the enlightenment of knowledge. The act of lighting the lamp signifies the dispelling of darkness and the illumination of wisdom and knowledge. The lamp-lighting ceremony was graciously led by the distinguished chief guests, Prof. Rumina Sethi, Dean of University Instruction, and Prof. Yogna Rawat, Director of the Research and Development Cell. They were accompanied by Dr. Anuj Sharma, Chairperson of DCSA, and Prof. Sonal Chawla, along with Prof. R.K. Singla, and other esteemed faculty, staff, and participants, whose presence honored and enriched the occasion.





After the lighting of the lamp, the session began with a heartfelt welcome speech by the Chairperson of DCSA and Coordinator of ICCAAIML, Dr. Anuj Sharma. In his address, he warmly welcomed the chief guests, dignitaries, faculty, staff, and participants, expressing his sincere gratitude for their presence and active involvement. He emphasized the importance of cultivating a vibrant research culture and fostering an environment that encourages innovation and collaboration. His words set a positive and inspiring tone for the event, motivating everyone to engage fully and contribute to the success of the conference.






Next, Prof. Sonal Chawla, the Coordinator of ICCAAIML, warmly welcomed the guests and motivated the participants with her encouraging words. She also highlighted the critical role of such conferences in promoting interdisciplinary research and collaboration. Following this, Prof. Rumina Sethi and Prof. Yojna Rawat, our esteemed Chief Guests, delivered insightful and inspiring speeches. They emphasized the responsible use of AI, noting that AI and ML play an integral role in our daily lives.

A major highlight of the inaugural session was the launch of the book "**Essentials of AI and Soft Computing**", authored by Dr. Anuj Sharma, Chairperson and Associate Professor at DCSA, and published by PHI Learning. This book is the culmination of years of dedication and expertise, seamlessly blending theoretical foundations with practical applications, making it an invaluable resource. It serves as a comprehensive guide for students pursuing bachelor's and master's degrees, including Ph.D. programs, as well as professionals eager to deepen their understanding of artificial intelligence and machine learning. The launch took place in the esteemed presence of the chief guest, distinguished participants, and faculty members, making the occasion even more memorable and adding to the honor of Dr. Anuj Sharma's remarkable achievement.



The keynote address was delivered by Dr. Satyajit Jena, a distinguished academician from IISER Mohali.In his insightful talk, titled "**Bridging Brains and Machines: The Legacy of Neural Networks in AI**", Dr. Jena explored the

transformative impact of Artificial Intelligence and Machine Learning. He discussed the pivotal role of neural networks in shaping the future of technology, highlighting the combination of Hopfield networks and Boltzmann machines as foundational techniques that have influenced the development of modern AI, including large language models (LLMs).Dr. Jena also explained how these advances are connected to the "Attention Is All You Need" paper, which introduced the Transformer architecture, an important development in LLMs. Additionally, he highlighted the immense potential of quantum computing, elaborating on its fundamental principles and offering valuable perspectives on the evolving landscape of these cutting-edge technologies. After an engaging session, the attendees took a break for tea and snacks, offering a moment for informal discussions and networking.

Following the break, Prof. Elisa Barney Smith from Luleå University of Technology (LTU), Sweden, introduced the innovative concept of "**Using AI and Machine Learning to further studies in Humanities**." Prof. Smith discussed the integration of AI with humanities research, emphasizing how this interdisciplinary approach can revolutionize traditional fields. In her talk, she covered the applications of AI and machine learning in various areas, including autonomous vehicles, ImageNet, ChatGPT, biomedical applications, as well as voice and face recognition. Next, she discussed paper fiber analysis, a technique used to determine the origin of paper and identify where it was manufactured. She also introduced digital humanities, a field that combines computing with the humanities to analyze and understand various disciplines within the humanities. She further addressed the challenges and complexities of applying AI in the humanities, offering valuable insights into how these technologies can contribute to the advancement and transformation of humanities studies in the future.

Post-lunch, the afternoon sessions were held across three venues: the CIL Hall at the CIL Building, RUSA Lab, and Lab 2 in the DCSA department. These sessions featured a series of conference presentations, covering a wide range of topics. The diversity of subjects sparked engaging academic



Dr. Satyajit Jena



Prof. Elisa Barney

discussions, encouraging the exchange of innovative ideas and perspectives. The sessions concluded at 5:00 PM, followed by tea and snacks, bringing the first day of the conference to a successful and productive close.

Day 2: Research Talks and Valedictory Session

The Conference began with renewed enthusiasm and anticipation, hosted exclusively at the RUSA Lab in the DCSA department. The day featured insightful talks from multiple speakers, each contributing valuable perspectives on digital infrastructure, AI, and data-driven solutions.

The opening session was presented by **Dr. Jyotirmoy Dutta** from IISc Bengaluru and CDPG, who spoke on "**Driving the Digital Public Infrastructure Growth Story with Data Exchanges.**" Dr. Dutta explored the transformative potential of data exchanges in accelerating India's Digital Public Infrastructure (DPI) growth, emphasizing the role of Aadhaar-linked services, DigiLocker, and agricultural data exchanges in enabling inclusive and efficient service delivery. He highlighted impactful initiatives such as the Telangana government's **Rythu Bandhu scheme**, which provides financial support to farmers, and **Namma Yatri**, a DPI-based platform.

Dr. Dutta also introduced **ADEX**, a cloud-based platform designed to overcome challenges like data silos and technical interoperability, stressing the importance of metadata catalogs and robust frameworks for authorization and resource management. He shared use cases including: **Vadodara Green Corridors, Varanasi Smart Waste Management (Smart Bins), Pune Forest Fire Management (Tensor**

Location Mapping), Public Transit Seat Availability, Smallholding Farmers Loan Facilitation. Dr. Dutta encouraged participants to explore such projects using platforms like **IUDX** (Indian Urban Data Exchange), accessible at iudx.org.in.

Following Dr. Dutta's session, **Dr. Balwinder Singh**, Joint Director at C-DAC Mohali, engaged the audience with his session on "**AI-Driven Solutions: Case Studies in Agriculture, Healthcare, and Rewards.**" His talk focused on the services offered by C-DAC in the fields of AI and Machine Learning, including honeypot security systems, e-governance tools, AI-powered virtual farm fencing, and automated hydroponics. He also highlighted **PARAM Smriti**, a high-performance supercomputing facility developed by **NABI, Mohali**, in collaboration with C-DAC, which supports advanced research in agri-food biotechnology.

Dr. Singh showcased several pioneering projects, including: **Automated Aquaponics for Fish and Crop Farming, Mobile Telemedicine Solutions, Cattle Identification Using Biometric Muzzle Patterns, Development of an Integrated Solution for Automatic Assessment of Autism (DISAAA)-Autism Assessment Tool , Electronic Nose for Soil Analysis, Organic Fertilization System for Crop Recommendation, AI-Based Diagnostic Platform for Polycystic Ovary Syndrome (PCOS), Karkat Nirnay Yantra for Breast Cancer, Chip for Single-Lead Wearable Electrocardiogram (ECG), AI-Based Virtual Farm Field Fencing System using YOLOv4, Intelligent Precision Farming Solution "e-Bagan".**

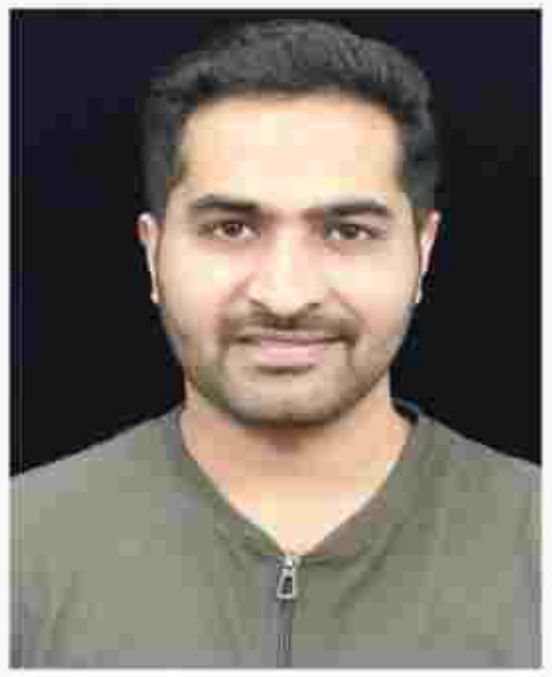


Dr. Jyotirmoy Dutta



Dr. Balwinder Singh

Following the tea and snack break, **Dr. Muhammad Suhil**, Applied Scientist II at Amazon Ads, captivated the audience with his talk on "**Efficient Fine-tuning of LLMs.**" His presentation focused on the challenges and future directions in Natural Language Processing (NLP), specifically around the fine-tuning of Large Language Models (LLMs). Dr. Suhil discussed the foundational role of transformers as building blocks in LLMs and explained key concepts like BERT, GPT, Prompt Engineering, and RAG (Retrieval-Augmented Generation). He also highlighted techniques such as PEFT (Parameter-Efficient Fine-Tuning), emphasizing the importance of KV cache management, Adapter Pruning, and Quantization for optimizing model performance. Dr. Suhil explained how these advanced fine-tuning strategies help overcome the evolving challenges in NLP applications, particularly by improving the efficiency and scalability of large models.



Dr Mahamad Suhil, Amazon Ads

After the lunch break, the **Valedictory Session** commenced with the conference coordinators, Prof. Sonal Chawla and Dr. Anuj Sharma, Chairperson of the Department, along with the participants, extending a warm welcome to the Chief Guest, **Prof. Y. P. Verma**, Registrar of Panjab University. Subsequently, Prof. Y. P. Verma, Prof. R. K. Singla, Prof. Sonal Chawla, and Dr. Anuj Sharma presented certificates to the participants in recognition of their significant contributions and active involvement in the conference.



Prof Y.P. Verma

Following this, Prof. Verma delivered a speech in which he appreciated the conference coordinators and participants for their unwavering commitment to advancing research and development and fostering innovation.

He also encouraged the participants to continue their efforts in pushing the boundaries of knowledge, contributing to future advancements, and making meaningful contributions to society.



The conference coordinators presented the report, expressing heartfelt gratitude to all colleagues, staff, and participants for their invaluable contributions. They applauded the meticulous planning and tireless efforts of the team, which were instrumental in ensuring the event's success. The session concluded with a tea and snacks break. The conference brought together nearly 100 participants, creating a dynamic environment for meaningful discussions and collaboration. The overwhelming enthusiasm and active engagement of participants made the event a great success, showcasing its positive impact and significance. As the conference concluded, it left a lasting impression, inspiring all to continue their pursuit of excellence and collective growth in research and development.

Article Written by:
S RATNA (Research Scholar)

About Faculty



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Chairperson



Prof. R. K. Singla



Prof. Indu Chhabra



Prof. Sonal Chawla



Prof. Anu Gupta



Dr. Jasleen Kaur Bains



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Mrs. Supreet Thapar



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